

Nodes and streams

The product balance and the water balance, step by step

Assignment “Product and Water Balances” · circuit-balancing computer lab · exam topics T9, T10, T11

One assignment, one lab and three exam topics are the same piece of arithmetic seen from different sides. Canvas calls it **Product and Water Balances** (the sheet ProductWaterBalances.pdf with the workbook ProductWaterBalances-1.xlsx); the Friday computer class calls it circuit balancing with HSC Sim; the written exam calls it T9, T10 and T11. This note works the whole thing end to end on the numbers the course itself uses, in the order that never gets stuck.

Read it with the workbook open. Every section ends with the check that tells you whether the sheet is right, because a balance that does not close is the only error message this arithmetic gives you.

The one rule underneath all of it

Draw a box. At steady state nothing accumulates inside it, so for every quantity that is conserved, what enters equals what leaves. Solids are conserved, each element is conserved, water is conserved. Particle size is not. Grades are conserved only if the minerals are not transformed.

1 · Nodes and streams: draw the box before you type

A **node** is any point where streams meet or split: a separator (one in, two out), a mixer (two in, one out), a machine. A **stream** is an arrow between two nodes, or between a node and the outside world. The input file of HSC Sim, and the left-hand column of the Excel sheet, are nothing but that numbering written down. Number the streams on paper first; the lab sheet asks for a simplified block diagram with numbered nodes and streams for exactly this reason.

Every node gives you equations, and you should count them before you start:

- one **solids** balance per node;
- one balance per **element** assayed, per node;
- one **water** balance per node, once you have the dilution ratios.

Choosing the box is the whole skill

Three boxes drawn on the same circuit give three different, equally true balances. Around the whole plant only the feed, the concentrate and the tail cross the line, and the recycle never appears. Around the mill alone the recycle is the biggest stream there is.



Box A, the whole circuit: product = F, and the recycle is invisible. Box B, the mill only: discharge = F + R. Same plant, two true statements.

In plain words

Choose the box so that the unknown you want crosses its edge, and as few other unknowns as possible. Around the whole circuit the recycle vanishes, which is what you want when you are after the products. Around one unit the recycle appears, which is what you want when you are after the circulating load.

Normal and non-normal nodes

A **normal** node splits one stream into two, or merges two into one, and carries compositional information. A mill has one feed and one discharge with the same composition, so it separates nothing: that is a **non-normal** node, and grinding circuits are full of them. They are not an error. They simply give you a solids balance and no useful element balance, so do not expect assays to pin the split there.

2 · The product balance

You weigh the feed and you assay everything, but you rarely weigh the products. The feed tonnage plus the assays are enough. With a feed of M_f tonnes and n products:

$$M_f = \sum M_i \quad \text{and} \quad M_f a_f = \sum M_i a_i$$

One solids balance, plus one element balance for every product beyond the first. In matrix form $A p = m$, where the first row of A is all ones and each further row holds one element's assays; the answer is $p = A^{-1} m$. The course's PRODUCT.XLS does this for up to five products, and the assignment template is the same machine with the columns already drawn.

Which element for which product

One element per product beyond the first, and the natural choice is each concentrate's own main metal: copper for the copper concentrate, zinc for the zinc concentrate. That is where the grade contrast between streams is largest, and contrast is what pins the split. An element with the same grade everywhere tells you nothing about the masses.

The two-product formula

When a unit or a whole plant makes only two products, the solids balance and one element balance solve in closed form. With grades f , c and t :

$$C / F = (f - t) / (c - t) \quad R = c (f - t) / [f (c - t)] \times 100$$

The concentrate mass is the feed times how far the feed grade sits between the tail and the concentrate. The recovery needs no tonnage at all, only three assays, which is why plants quote it every shift. Its weak point is the denominator $c - t$: with little contrast, a small assay error moves the answer a long way.

The three numbers the worksheet asks for at the end

- **Recovery R** — the share of a metal in the feed that reached its own concentrate, $R = M_c c / (M_f f) \times 100$.
- **Concentrator difference** — metal in minus metal out. For the element whose assays sized the streams it is zero **by construction**; for every other element it is a real number, and it measures how well sampling and assaying hang together.
- **Concentrator loss** — metal that did not reach its own concentrate, per tonne of feed. Two definitions circulate: the final tail only, or the tail plus what was misplaced in the other concentrate. The examiner uses the second. **Say which one you used.**

3 · Worked: recoveries given (Pb-Zn-Ag, 110 000 t/month)

Selective flotation: lead first, then zinc from the lead-circuit tail. Four streams cross the worksheet. Feed 110 000 t/month at 1.50 % Pb, 8.00 % Zn, 30 g/t Ag. Lead cake 70.0 % Pb at R = 90.5 %, 890 g/t Ag. Zinc cake 55.0 % Zn at R = 96 %, 50 g/t Ag. Final tail 0.10 % Pb, 0.30 % Zn, 5 g/t Ag.



Lead cake leaves the first node, zinc cake the second; the lead-circuit tail (mpPb) is the internal stream that feeds zinc flotation.

Step 1. Metal in the feed. Mass times grade: 110 000 × 1.50 % = 1 650 t Pb, × 8.00 % = 8 800 t Zn, × 30 g/t = 3 300 kg Ag.

Step 2. Concentrate masses from the recoveries. A recovery fixes the metal in the concentrate; the grade then fixes the mass.

$$m_{\text{Pb cake}} = 0.905 \times 1\,650 / 0.700 = 1\,493.25 / 0.700 = \mathbf{2\,133.2\,t}$$

$$m_{\text{Zn cake}} = 0.96 \times 8\,800 / 0.550 = 8\,448 / 0.550 = \mathbf{15\,360.0\,t}$$

Step 3. The intermediate stream and the tail, by difference. mpPb = 110 000 – 2 133.2 = 107 866.8 t; final tail = 107 866.8 – 15 360.0 = **92 506.8 t**.

Step 4. Fill the metal columns. The tail grades are given: 92 506.8 × 0.10 % = 92.51 t Pb, × 0.30 % = 277.52 t Zn, × 5 g/t = 462.5 kg Ag. The grades that are **not** given come out by difference, so the two elements whose recoveries you used close exactly:

$$\text{Zn in the Pb cake} = 8\,800 - 8\,448 - 277.52 = 74.48\,\text{t} \Rightarrow 74.48 / 2\,133.2 = 3.49\,\% \text{ Zn}$$

$$\text{Pb in the Zn cake} = 1\,650 - 1\,493.25 - 92.51 = 64.24\,\text{t} \Rightarrow 64.24 / 15\,360 = 0.42\,\% \text{ Pb}$$

Step 5. Silver, the by-product. Its grades are given everywhere, so it does **not** close. Pb cake 2 133.2 × 890 = 1 898.6 kg; Zn cake 15 360 × 50 = 768.0 kg; tail 462.5 kg; sum 3 129.1 kg against 3 300 kg fed. Concentrator difference = –170.9 kg, about 2 g/t of feed: the assays do not account for 5 % of the silver.

Step 6. Closing numbers. Difference = 0 for Pb and Zn by construction. Loss, examiner's definition: Pb (92.51 + 64.24) / 110 000 = 1.43 kg/t; Zn (277.52 + 74.48) / 110 000 = 3.20 kg/t. Counting the final tail alone gives 0.84 and 2.52 kg/t.

Product	Dry weight t	Ag g/t	Ag kg	Pb %	Pb t	Zn %	Zn t
Feed ore	110 000.0	30	3 300.0	1.50	1 650.0	8.00	8 800.0
Pb cake	2 133.2	890	1 898.6	70.00	1 493.3	3.49	74.5
Zn cake	15 360.0	50	768.0	0.42	64.2	55.00	8 448.0
Final tail	92 506.8	5	462.5	0.10	92.5	0.30	277.5
Σ products	110 000.0	28	3 129.1	1.50	1 650.0	8.00	8 800.0
Concentrator difference		–2	–170.9	0.00	0.0	0.00	0.0

Silver distribution: 60.7 % to the lead cake, 24.5 % to the zinc cake. Concentrator loss 11.19 g/t Ag when the difference is counted as lost metal.

4 · Worked: only grades given (Aitik, 92 500 t/day)

No recoveries this time, only grades, and three elements. The masses come from the two-product formula applied twice, on copper, the element with the biggest contrast. Gold and sulphur then ride along on those masses, and their concentrator differences are real numbers, not zero. The scavenger concentrate returns to the rougher and never leaves, so it does not appear in the products balance at all.

Step 1. Rougher-scavenger section, two-product formula on copper. Feed 0.28 % Cu, rougher concentrate 12.0 %, scavenger tail 0.022 %:

$$M_{\text{rougher conc}} = 92\,500 \times (0.28 - 0.022) / (12.0 - 0.022) = 92\,500 \times 0.258 / 11.978 = \mathbf{1\,992.4\,t}$$

Scavenger tail = 92 500 – 1 992.4 = 90 507.6 t.

Step 2. Cleaner section, the same formula again. Its feed is the rougher concentrate (12.0 %); products final concentrate (29.0 %) and cleaner tail (0.39 %):

$$M_{\text{final conc}} = 1\,992.4 \times (12.0 - 0.39) / (29.0 - 0.39) = 1\,992.4 \times 11.61 / 28.61 = \mathbf{808.5\,t}$$

Cleaner tail = 1 992.4 – 808.5 = 1 183.9 t.

Step 3. Metal columns from the grades: every mass times every grade, gold in g/t to kg, copper and sulphur in t.

Product	Dry mass t	Au g/t	Au kg	Cu %	Cu t	S %	S t
Feed ore	92 500.0	0.30	27.75	0.28	259.00	1.20	1 110.0
Final conc	808.5	9.2	7.44	29.0	234.47	31.0	250.6
Cleaner tail	1 183.9	2.4	2.84	0.39	4.62	1.10	13.0
Scavenger tail	90 507.6	0.10	9.05	0.022	19.91	0.90	814.6
Σ products	92 500.0		19.33		259.00		1 078.2
Concentrator difference			8.42		0.00		31.8

Reading the differences. Copper closes at zero because its grades sized the streams. Gold is short by 8.42 kg, 30 % of what was fed: a feed assay of 0.3 g/t sits near the detection limit and does not hang together with copper-based masses. Sulphur is short by 31.8 t, 2.9 %, an ordinary assay mismatch. Recoveries: Cu 90.5 %, Au 26.8 %, S 22.6 %. Losses: Cu 0.265 kg/t, Au 0.129 g/t, S 8.95 kg/t.

5 · The water balance

Slurry streams are described by the **dilution ratio**, the mass of water per mass of solids. One cubic metre of water is one tonne, so litres of water per kilogram of solids is the same number. Everything else is a rearrangement of it:

$$D = W / M \quad W = D \cdot M \quad F_w = 100 / (1 + D) \quad D = 100 / F_w - 1$$

F_w is the weight per cent solids. The pulp density a Marcy scale reads, with water at 1 and solids at ρ_s :

$$\rho_p = (D + 1) / (D + 1/\rho_s) \quad D = (\rho_s - \rho_p) / [\rho_s (\rho_p - 1)]$$

D is an exchange rate

It converts the tonnes you know into the water you want. $D = 2$ is one third solids by weight. A cyclone underflow at $D = 0.4$ is thick pulp. A thickener overflow at $D = 36$ is almost clear water. A dry feed has $D = 0$.

The order that never gets stuck

Every water-balance question in this course yields to the same five moves, in this order. Write each balance as an equation before you put numbers in it; the examiner says so on the cover.

- **Solids around the whole circuit.** At steady state what goes in comes out. Find the product streams first; the recycle is invisible from here.
- **Solids around each unit.** Now the recycle appears, one unit at a time, by difference.
- **Circulating load.** The solids that return to the mill divided by the fresh feed, times 100.
- **Water around the whole circuit.** All the fresh water added leaves with the products. That single equation usually gives the first unknown D.
- **Water around each unit,** then **check** on a box you have not used yet. If the two sides agree, all your answers are consistent.

$$CL = M_{\text{recycle}} / M_{\text{fresh feed}} \times 100$$

A circulating load of 200 % means the mill discharges three tonnes for every fresh tonne fed. It is always the last line of the question, because the recycle tonnage is what the water balance was hunting for.

6 · Worked: the pilot circuit (mill, cyclone, Wright Impact Tray)

Feed 14.3 kg/min of dry ore. Mill water 8.5 l/min. Cyclone underflow 29.5 kg/min at $D = 0.37$. WIT concentrate 3.2 kg/min at $D = 0.92$, with 10.7 and 9.0 l/min added on the two decks. Asked: the dilution ratios of the cyclone overflow, the cyclone feed and the WIT tailings, and the circulating load.



Cyclone overflow and WIT concentrate are the only streams that leave. The WIT tailings go back to the mill: that is the recycle.

Step 1. Solids around the whole circuit. Only two streams leave: overflow = $14.3 - 3.2 = 11.1$ kg/min.

Step 2. Solids around each unit. Cyclone: feed = $29.5 + 11.1 = 40.6$ kg/min. WIT: it receives the underflow (29.5) and splits it into concentrate (3.2) and tailings, $29.5 - 3.2 = 26.3$ kg/min back to the mill.

Step 3. Circulating load.

$$CL = 26.3 / 14.3 \times 100 = 184 \%$$

Step 4. Water around the whole circuit. All the fresh water, $8.5 + 9.0 + 10.7 = 28.2$ l/min, leaves with the two products:

$$W_{\text{overflow}} = 28.2 - 3.2 \times 0.92 = 25.26 \text{ l/min} \Rightarrow D_{\text{overflow}} = 25.26 / 11.1 = 2.28$$

Step 5. Water around the cyclone. Its feed water is the sum of the two outlets:

$$W_{\text{cyclone feed}} = 29.5 \times 0.37 + 25.26 = 36.17 \Rightarrow D_{\text{cyclone feed}} = 36.17 / 40.6 = 0.89$$

Step 6. Water around the WIT. In: the underflow water plus the two additions. Out: concentrate and tailings.

$$W_{\text{tailings}} = 10.92 + 9.0 + 10.7 - 2.94 = 27.67 \Rightarrow D_{\text{tailings}} = 27.67 / 26.3 = 1.05$$

Step 7. Close the loop, on a box you have not used. Mill water in = 8.5 fresh + 27.67 from the tailings = 36.17, which is the cyclone feed water found in step 5. The balance closes, so the three answers are consistent.

Stream	Solids kg/min	Water l/min	D
Feed ore	14.3	0	0
Cyclone feed	40.6	36.2	0.89
Cyclone underflow	29.5	10.9	0.37
Cyclone overflow (product)	11.1	25.3	2.28
WIT concentrate (product)	3.2	2.9	0.92
WIT tailings (to mill)	26.3	27.7	1.05

The 2020 and 2022 papers are the same circuit with 15.8 kg/min of feed and 10.0 l/min of mill water; everything else is identical.

7 · Worked: the three-mill grinding circuit

The hardest of the water balances and the most like a real plant: a rod mill, two ball mills, two cyclones, water added at the feed and at the discharge of every mill. The table gives a dilution ratio for every pumped stream, so each mill can be balanced on its own once its solids are known.

Read the table before you trust it

The row “Pump, 0.76” is the rod-mill discharge as it is pumped. “Feed cyclone 2, 0.84” is that same pulp after the ball-mill-2 discharge has joined it. Only with this reading are there as many equations as unknowns. State the reading in your answer: interpreting the flowsheet is part of the marks.

Feed 52.5 t/h at 4.57 % moisture. Moisture is on the **wet** mass, so the water in the feed is $52.5 \times 4.57 / 95.43 = 2.51 \text{ m}^3/\text{h}$.

Step c. Solids to each cyclone first; this is the backbone. Cyclone 2 feed = rod-mill discharge 52.5 + ball-mill-2 discharge 31.2 = 83.7 t/h, and its overflow is 52.5 t/h, the fresh feed again. Cyclone 3 feed = 52.5 + 24.1 = 76.6 t/h, overflow 52.5 t/h to flotation. Nothing accumulates.

Step a. Water added at each mill feed, working along the pulp.

- **Rod mill.** Discharge water = $52.5 \times 0.76 = 39.90$. In: 2.51 moisture + x_1 + 12.9 added at the discharge. So $x_1 = 39.90 - 2.51 - 12.9 = \mathbf{24.5 \text{ m}^3/\text{h}}$.
- **Ball mill 2.** Cyclone 2 feed water = $83.7 \times 0.84 = 70.31$, so ball mill 2 brings $70.31 - 39.90 = 30.41$. In: underflow water $31.2 \times 0.38 = 11.86$, plus x_2 , plus 5.9. So $x_2 = 30.41 - 11.86 - 5.9 = \mathbf{12.7 \text{ m}^3/\text{h}}$.
- **Ball mill 3.** Cyclone 2 overflow water = $70.31 - 11.86 = 58.45$. Cyclone 3 feed water = $76.6 \times 1.04 = 79.66$, so ball mill 3 brings 21.21. In: underflow $24.1 \times 0.57 = 13.74$ plus x_3 . So $x_3 = \mathbf{7.5 \text{ m}^3/\text{h}}$.

Step b. Dilution ratio of the flotation feed. Cyclone 3 overflow water = $79.66 - 13.74 = 65.93 \text{ m}^3/\text{h}$ with 52.5 t/h:

$$D_{\text{flotation feed}} = 65.93 / 52.5 = \mathbf{1.26 \text{ (44 \% solids)}}$$

Check. Total water added = $2.51 + 24.5 + 12.9 + 12.7 + 5.9 + 7.5 = 65.9 \text{ m}^3/\text{h}$, exactly the water leaving with the flotation feed. Closed.

Stream	Solids t/h	Water m ³ /h	D
Feed ore	52.5	2.5	0.05
Rod mill discharge (pump)	52.5	39.9	0.76
Cyclone 2 feed	83.7	70.3	0.84
Cyclone 2 underflow	31.2	11.9	0.38
Cyclone 2 overflow	52.5	58.5	1.11
Cyclone 3 feed	76.6	79.7	1.04
Cyclone 3 underflow	24.1	13.7	0.57
To flotation	52.5	65.9	1.26

8 · Driving the workbook

The assignment template (ProductWaterBalances-1.xlsx) and the course's PRODUCT.XLS are the same worksheet: one row per stream, one pair of columns per element (grade, then metal), and the closing rows at the bottom. Fill it in the order of sections 3 and 4, not left to right.

Column	What goes in it	How
Product	one row per stream that crosses the box	feed first, then each concentrate, then the tail
Dry weight	the tonnage of the stream	given for the feed; from recoveries or the two-product formula for the concentrates; the rest by difference
Grade, per element	% or g/t	typed where the sheet gives it, computed by difference where it does not
Metal, per element	t or kg	= dry weight × grade, with the unit conversion; this is the column that must add up
Distribution %	share of the feed metal in that stream	= metal in the stream ÷ metal in the feed × 100; the column sums to 100
Σ products	sum of every stream except the feed	compare it with the feed row: that comparison is the whole point
Concentrator difference	feed metal – Σ products metal	zero by construction for the element that sized the streams
Concentrator loss	metal that missed its own concentrate, per t of feed	state whether you counted the tail only or the tail plus the misplaced metal

Three habits that save the sheet

- **Keep the units in one system.** Per cent next to per cent, g/t next to g/t. Mixing them is the single commonest way the metal column stops adding up, and in a least-squares balance it also wrecks the coefficients of A.
- **Never type a number the sheet can compute.** Every grade that came out by difference should be a formula pointing at the metal column, so that when you correct one assay the whole worksheet follows.
- **Put the check in a cell.** A cell holding Σ products – feed, for every element, next to a cell holding total water in – total water out. Watch those two cells while you work, not the pretty table.

9 · What goes wrong, and what it means

Symptom	What it is telling you
A negative tonnage after balancing	The equations are wrong, not the plant. Check which streams you declared internal and whether a node exists that you did not draw.
A stream balanced to zero flow	Same cause. Usually a node connected to the wrong stream number.
An adjustment larger than three standard deviations	Either the measurement is worse than you claimed, or the flowsheet is wrong: a stream forgotten, a node that does not exist.
The difference is large for one element only	That element's assays do not hang together with the masses. Expected for gold at grams per tonne; suspicious for copper.
The answer swings when one assay moves slightly	You balanced on an element with little contrast. Use the concentrate's own main metal, where $c - t$ is large.
Weight % solids entered as a component	It is not conserved on its own. Convert it to a water flow or a dilution ratio first, then let water be the component.
A relative standard deviation above 1.0	The error is larger than the value: the measurement carries no information and the fit will move it anywhere. Good data sit at a few per cent.

What weighting does and does not do

Weights do not make the tonnages more accurate. They decide **where the disagreement is dumped**: on the measurements you told the program to distrust. You cannot force a balance out of bad data; somewhere the error has to surface, sometimes as an absurd adjustment. The compendium's rule of thumb for relative standard deviations: measured mass flows 1 %; assays 1 % when very good, 10 % good, 30 % fair, 50 % bad, 100 % hopeless.

10 · The six statements (T9)

One exam question is six true-or-false statements about this material, one point each, and the point is only safe when you can say why.

Statement		Why
A node is a connection point in a flowsheet.	True	Any point where streams meet or split: a separator, a mixer, a machine. Balances are written around nodes.
A non-normal node can be found in a grinding circuit.	True	A mill has one feed and one discharge of the same composition, and a mill-plus-cyclone loop reduces to nodes with more than two products. Both are non-normal and typical of grinding.
Analyses of gold are uncertain.	True	Grades are grams per tonne and the grains are coarse and rare, so sampling and assay errors are large. Gold gets a big standard deviation or is left out of the fit.
It is best to use analyses with large contrast.	True	An element whose grade differs strongly between products pins the split. One with the same grade everywhere says nothing about the masses.
Weight % solids can be used directly in MATBAL.	False	It is not conserved on its own. Turn it into a water flow or a dilution ratio, or give solids and pulp masses separately.
A relative standard deviation above 1.0 is good.	False	The error exceeds the value, so the measurement carries no information. Good data are a few per cent.

Where this returns

- **The assignment.** Product and water balances, due in Canvas; the exercise session walked the same sheet.
- **The computer lab.** The same arithmetic with more measurements than unknowns, solved by weighted least squares in HSC Sim; the report is graded inside examination module 0003.
- **The written exam.** T9 as the six statements, T10 and T11 at 10 points each. A full-mark answer has three parts: the box you chose with its streams named, the equations in symbols, and the numbers with units plus a check that closes.